



This guide covers the five non-obvious steps required to build this Notion template correctly. Read each section **before** opening the Notion interface.

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1 The Strict Database Creation Sequence

Notion requires the **target database to exist before a relation pointing to it can be created**. Building in the wrong order means you cannot set up the relation — Notion simply will not offer the target as a link option. Follow this exact sequence:

#	Database	Why This Position
1	Port Forwarding Rules	No outbound relations. Network Map relates to it — target must exist first.
2	Network Map	Relates to Port Forwarding Rules (already built). Hardware relates to it — target must exist first.
3	Hardware Deployment	Relates to Network Map (already built). Central database in the schema.
4	Services & Container Registry	Relates to Hardware Deployment (already built).
5	Storage Inventory	Relates to Hardware Deployment (already built).
6	Maintenance Log	Relates to Hardware Deployment (already built). Build last.

Tip: Open six Notion browser tabs before you start — one blank database per tab, created in order. Complete each database fully before moving to the next.

2 The One-Way Relation: Hardware → Network Map

By default, Notion creates a **two-way relation**: when you add a relation from Hardware Deployment to Network Map, Notion automatically adds a matching back-relation property inside Network Map. The Network Map is a clean IP registry. A hardware back-reference column clutters it. Delete it.

- 1 Open **Hardware Deployment** → **+ Add a property** → select **Relation**.
- 2 When prompted for the target database, select **Network Map**.
- 3 Name this property exactly: **Network Entries** (case-sensitive — referenced by the Total Exposed Ports rollup).
- 4 Complete the dialog. Notion creates a back-relation in Network Map automatically.
- 5 Open **Network Map** → click the **Properties** icon (top-right of any table view).
- 6 Find the auto-created property. Notion names it after the source database (e.g., “Hardware Deployment”).

7 **Delete it entirely.** Do not hide — delete. A hidden property still surfaces in filtered views and full-page record opens.

Result: Hardware Deployment records show their linked Network Map entries. Network Map records show no hardware back-reference. The IP registry stays clean.

3 The Two-Hop Rollup Dependency

Notion does not traverse two relation hops in a single rollup. To count exposed ports per hardware node, build an intermediate rollup inside Network Map first, then roll *that* up from Hardware Deployment. **These two rollups must be created in sequence.**

Step A — Intermediate rollup in Network Map:

Field	Value
Property name	Open Port Count
Property type	Rollup
Relation	Port Rules (→ Port Forwarding Rules)
Property to roll up	Rule Name
Aggregation	Count All

Step B — Parent rollup in Hardware Deployment:

Field	Value
Property name	Total Exposed Ports
Property type	Rollup
Relation	Network Entries (→ Network Map)
Property to roll up	Open Port Count
Aggregation	Sum — NOT Count All (see note below)

Why Sum and not Count All: “Count All” counts the number of linked Network Map records (i.e., the number of network interfaces). “Sum” adds up the *Open Port Count* values across all those interfaces, giving you the

true total number of port rules exposed by that hardware node.

4

Case-Sensitivity & Emoji Filter Rules

The Notion Formula 2.0 `prop()` function resolves property names as literal strings. One wrong character — a lowercase letter, a missing space, a missing parenthesis in a property name — causes the formula to return a blank value silently, with no visible error.

Database	Property Name (exact)	Referenced By
Hardware Deployment	Maintenance Mode	Status Flag
Hardware Deployment	Last Seen	Status Flag, Days Since Last Seen
Hardware Deployment	Warranty Expiry	Warranty Status
Hardware Deployment	Network Entries	Total Exposed Ports rollup
Storage Inventory	Capacity (TB)	Storage % Used
Storage Inventory	Used (TB)	Storage % Used
Port Forwarding Rules	External Port	Port Rule Label
Port Forwarding Rules	Internal Port	Port Rule Label
Port Forwarding Rules	Protocol	Port Rule Label
Network Map	Open Port Count	Total Exposed Ports rollup

Emoji filter rule — use Contains, never Equals:

The Status Flag formula returns strings with leading emoji characters (e.g., `Active`, `Offline`, `Stale`). Notion's filter engine does not reliably match emoji with the **Equals** operator. All Status Flag view filters must use **Contains**.

Operator	Filter Value	Result
Equals	"🟩 Active"	BROKEN — emoji matching unreliable
Contains	"Active"	CORRECT
Contains	"Offline"	CORRECT
Contains	"Maintenance"	CORRECT

Contains "Stale"

CORRECT

5 Stress-Test With Real Hardware Data

Before packaging the template for sale, validate every relation, rollup, and formula with actual data. The two test records below cover the two most architecturally distinct use cases this template is built for: legacy x86 hardware repurposed with a new OS, and an ARM SBC running a container stack.

Test Record 1 — Legacy Hardware Repurpose (iMac + Batocera):

Property	Value
Node Name	imac-2011-batocera
Node Type	Physical Machine
OS / Runtime	Batocera
Architecture	x86_64
CPU Model	Intel Core i5-2500S
RAM (GB)	8
Role	Emulation
OS Lifecycle State	Production
Maintenance Mode	Unchecked
Last Seen	Today's date
Network Entries	Link to a new Network Map record: imac-2011 / 192.168.1.x

Test Record 2 — SBC Container Stack (Raspberry Pi 4):

Property	Value
Node Name	rpi4-container-stack
Node Type	SBC

OS / Runtime	Raspberry Pi OS
Architecture	ARM64
CPU Model	Broadcom BCM2711
RAM (GB)	4
Role	DNS / Ad Blocker, Reverse Proxy, Monitoring
OS Lifecycle State	Production
Maintenance Mode	Unchecked
Last Seen	Today's date
Network Entries	Link to a new Network Map record: rpi4 / 192.168.1.y

Validation checklist:

Status Flag	Both records show "Active" immediately after setting Last Seen to today. Toggle Maintenance Mode on one — it must switch to "Maintenance" regardless of Last Seen.
Days Since Last Seen	Shows "0d ago" on both records when Last Seen is today.
One-Way Confirm	Open each Network Map record. There must be no "Hardware Deployment" column or any back-relation property visible anywhere in the record.
Two-Hop Rollup	Create a Port Forwarding Rule and link it to the RPi4 Network Map entry. The RPi4 Hardware Deployment record must show 1 in Total Exposed Ports. Add a second rule — it must increment to 2.
Warranty Status	Set Warranty Expiry to 30 days from today on the iMac record. It must show "Expiring Soon". Set it to a past date — it must show "Expired".
Formula 4	Create a Storage Inventory record linked to the RPi4. Enter Capacity = 0 — the % Used formula must return "—" (not a division error).

Appendix A — Formula Quick Reference

Paste each formula verbatim into the Notion formula property editor. Property names are case-sensitive. Do not rename any property after formulas are in place.

Formula 1 — Status Flag

Database: Hardware Deployment | Output: Text

FORMULA 1 — STATUS FLAG (add emoji prefixes as desired)

```
if(
  prop("Maintenance Mode"),
  "Maintenance",
  if(
    empty(prop("Last Seen")),
    "Unknown",
    if(
      dateBetween(now(), prop("Last Seen"), "days") > 7,
      "Offline",
      if(
        dateBetween(now(), prop("Last Seen"), "days") > 1,
        "Stale",
        "Active"
      )
    )
  )
)
```

Formula 2 — Days Since Last Seen

Database: Hardware Deployment | Output: Text

FORMULA 2 — DAYS SINCE LAST SEEN

```
if(
  empty(prop("Last Seen")),
  "_",
  format(dateBetween(now(), prop("Last Seen"), "days")) + "d ago"
)
```

Formula 3 — Warranty Status

Database: Hardware Deployment | Output: Text | `dateBetween(Expiry, Now) = Expiry - Now`

FORMULA 3 — WARRANTY STATUS

```
# Negative result = warranty has expired
if(
  empty(prop("Warranty Expiry")),
  "_",
  if(
    dateBetween(prop("Warranty Expiry"), now(), "days") < 0,
    "Expired",
    if(
      dateBetween(prop("Warranty Expiry"), now(), "days") < 90,
      "Expiring Soon",
      "Under Warranty"
    )
  )
)
```

Formula 4 — Storage % Used

Database: Storage Inventory | Output: Text (switch to numeric variant for rollup aggregation)

FORMULA 4 — STORAGE % USED

```
if(
  empty(prop("Capacity (TB)")),
  "_",
  if(
    prop("Capacity (TB)") == 0,
    "_",
    format(round((prop("Used (TB)") / prop("Capacity (TB)")) * 100)) + "%"
  )
)
```

Formula 5 — Port Rule Label

Database: Port Forwarding Rules | Output: Text

FORMULA 5 — PORT RULE LABEL

```
format(prop("External Port")) + " → " + format(prop("Internal Port")) + " / " + prop("Protocol")
```

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